MIL-M-13355A(MU)
16 June 1965
SUPERSEDING
MIL-M-13355(Ord)
30 March 1954

MILITARY SPECIFICATION

MOUNT, TELESCOPE: M3A1

1. SCOPE

1.1 This specification covers performance requirements, related quality assurance provisions, and packaging data for a mount which incorporates an elevation quadrant and a mounting surface for an optical sighting instrument (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army Munitions Command

A7597736	Mount, Telescope: M3A1
D10689	Adapter
D23553	Elevation and Cross-leveling Testing Fixture

PACKAGING DATA SHEET

MIL-P-14232/P7597736	Packaging of Mount, Telescope: M3A1

FSC 1240

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(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The mount shall be manufactured in accordance with Drawing A7597736 and drawings pertaining thereto.

3.2 General specification.- The contractor shall be responsible for adherence to the following requirements of MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.3 Environmental service conditions.

3.3.1 Extreme operating temperatures.- The mount shall meet the applicable requirements of 3.6 while exposed to, and thermally stabilized at, -40° and +150° Fahrenheit (F).

3.3.2 Vibration.- The mount shall show no evidence of physical failure due to being vibrated at a frequency of 30 ± 5 cycles per second (cps) with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes ± 15 seconds at a standard ambient temperature between +60° and +90°F. Subsequent to vibration the mount shall meet the following requirements of this specification.

3.4 Orientation.- The following requirements shall be met when the mount is positioned with the mounting shaft vertical and the telescope locating surface in a horizontal plane:

(a) The elevation scale shall be at zero index.
(b) The elevation micrometer shall be at zero index.
(c) The cross level adjustment shall be at mid-point of travel within 10 mils.
(d) The cross level vial shall be centered.
(e) The angle of site scale shall be at "3" index.
(f) The angle of site micrometer shall be at zero index.
(g) The angle of site level vial shall be centered within the thickness of a vial graduation line.

3.5 Accuracy.

3.5.1 Elevation error.- Error in the elevation mechanism shall not exceed 1.0 mil at any point on the scale (0 to 1200 mils) including backlash.

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3.5.2 Angle of site error.- Error in the angle of site mechanism shall not exceed 1.0 mil at any point on the scale (0 to 600 mils) including backlash.

3.5.3 Backlash.

(a) Backlash in the elevation mechanism shall not exceed 0.5 mil.
(b) Backlash in the angle of site mechanism shall not exceed 0.5 mil.
(c) Backlash in the cross level mechanism shall not exceed 1.5 mil.

3.5.4 Plumb travel.- From the position established in 3.4, a line of sight, parallel to the telescope locating surface and perpendicular to the plane of the cross level vial locating holes, shall not deviate from a plumb line by more than 0.5 mil (total excursion 0.5 mil) when the elevation mechanism of the mount is actuated through an excursion of 1200 mils.

3.5.5 Cant.- From the position established in 3.4, but with the mounting shaft inclined 7 degrees to the right of vertical, a line of sight parallel to the telescope locating surface and perpendicular to the plane of the cross level vial locating holes shall not deviate from a plumb line by more than 0.5 mil (total excursion 0.5 mil) when the elevation mechanism is actuated through an excursion of 1200 mils. This requirement shall also apply with the mounting support inclined 7 degrees to the left of vertical.

3.6 Operability.- The running torque required to rotate the adjustments of the mount shall not exceed the values specified at temperatures indicated in Table I.

TABLE I - Operability

Control	+60° to 90°F	-40° and +150°F
Elevation knob	50 in-oz.	5 in-lb
Angle of site knob	30 in-oz.	60 in-oz.
Cross level knob	60 in-oz.	5 in-lb

3.6.1 With the elevation lock secured, the elevation knob shall remain immovable when torqued to 30 in-lb.

3.7 Azimuth correction.- With the mount positioned initially as specified in 3.4, and with a line of sight established parallel to the telescope locating surface and perpendicular to the plane of the cross level vial locating holes, measured angles of azimuth correction shall be as indicated in Table II when the designated cant and elevation angles are introduced.

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TABLE II - Azimuth Correction

Angle of Cant	Elevation Angle	Azimuth Correction	Azimuth Correction Tolerance
7 degrees (124.4 mils)	500 mils	66.9 mils	± 1.0 mil
7 degrees	800 mils	125.4 mils	± 1.5 mils
7 degrees	1100 mils	236.1 mils	± 2.0 mils

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with MIL-STD-105, and the inspection level and sampling plans specified in Table III titled "Classification of Defects". Examinations and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

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TABLE III - CLASSIFICATION OF DEFECTS

Use Inspection Level II of Table I with
Sampling Plan Table II-A of MIL-STD-105

		REQUIREMENTS	TEST PROCEDURES
CRITICAL: NONE			
MAJOR: AQL 1.0 percent defective			
101.	Vibration	3.3.2	4.5.2
102.	Azimuth correction	3.7	4.5.4
103.	Plumb travel	3.5.4	4.5.10
104.	Elevation error	3.5.1	4.5.5
105.	Angle of site error	3.5.2	4.5.7
106.	Backlash, elevation mechanism	3.5.3(a)	4.5.6
107.	Backlash, angle of site mechanism	3.5.3(b)	4.5.8
108.	Backlash, cross level mechanism	3.5.3(c)	4.5.9
109.	Cant	3.5.5	4.5.11
110.	Operability (60° to 90°F)	3.6, 3.6.1	4.5.12, 4.5.12.1
MINOR: NONE			

4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3 Special sampling.

4.3.1 General.- One mount shall be selected at random by a Government representative as a special sample from each 100 produced. The sample shall meet the requirements and tests in Table IV.

TABLE IV

	CHARACTERISTICS	REQUIREMENTS	TEST PROCEDURES
301.	Fabrication	3.1	Applicable drawings - visual
302.	General specification	3.2	MIL-F-13926 - visual

4.3.2 Environmental.- Three units shall be selected at random by a Government representative as a special sample from each 50 produced or from each month's production, whichever occurs first. The samples shall have met the requirements and tests in Table III and shall then meet the requirement and test in Table V.

TABLE V

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Operating temperatures, knob torque (-40° and +150°F)	3.3.1	4.5.1

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4.3.3 Failure of sample.- Should any one item of a standard sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.2.3).

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.4.2 Contractor furnished inspection equipment.

4.4.2.1 Government design.- All inspection equipment required by drawings forming a part of the contract and not provided by the Government shall be supplied by the contractor in accordance with the requirements of 4.4.

4.4.2.2 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test methods and procedures" specified in 4.5 and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5 Test methods and procedures.

4.5.1 Extreme operating temperatures.- Test equipment utilized in the performance of this test shall conform to the "Test Facilities" requirements of MIL-F-13926. The mount shall be exposed to the temperature conditions of 3.3.1. As a precautionary measure, the temperatures shall be changed gradually to avoid thermal shock. When thermal stabilization is reached at each temperature extreme, the mount shall be tested in accordance with 4.5.12 to determine compliance with 3.3.1. The temperature shall then be reduced to a standard ambient temperature between 60° and 90°F.

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4.5.2 Vibration.- Standard vibrating equipment, capable of providing the amplitude and frequency specified in 3.3.2, shall be utilized for this test. Unless otherwise specified, the accuracy of the applicable test equipment and test equipment measurements shall conform to the "Test Facilities" requirements of MIL-F-13926. The mount shall be secured on the vibration table in its operating position, using suitable mounting means. The mount shall be vibrated in a vertical plane with the motion and frequency applied for 5 minutes ± 15 seconds. At the conclusion of this vibration, the mount shall be subjected to a visual and tactile examination for any evidence of physical failure. The mount shall then be subjected to the following tests in compliance with 3.3.2.

4.5.3 Orientation.- Orientation shall be performed utilizing special testing equipment to be compatible with the Elevation and Cross-Leveling Testing Fixture, Drawing D23553, in conjunction with Adapter, Drawing D10689, and an auxiliary telescope with a reticle containing vertical and horizontal cross lines. The elevation and cross leveling fixture, with adapter attached, shall be cross leveled prior to attaching the mount under test. With the mount positioned on the fixture, position the auxiliary telescope on the telescope locating surface. All scales, micrometer settings and level vial bubbles shall then be oriented to the settings specified in 3.4(a) through 3.4(g) to conform with 3.4.

4.5.4 Azimuth correction.- The test shall be performed utilizing the elevation test fixture with mounting adapter and a 3-power auxiliary telescope. Clamp the telescope on the top machined surface of the adapter fixture at 90 degrees to the axis about which the test fixture rotates in elevation. Sight through the telescope and establish a horizontal and vertical line target. Remove the telescope, and secure the mount to the adapter fixture. Position the auxiliary telescope on the telescope mounting surface. The mount shall be positioned as specified in 3.7 and 4.5.3. From the zero position, direct the telescope to bring into coincidence the vertical line of the target with the vertical reticle line. Two additional vertical lines (plumb lines) one to the left and one to the right of the established vertical line shall be provided. The right and left plumb lines will compensate for displacement of the vertical center line of the telescope's locating surface when the test fixture is rotated 7 degrees right and 7 degrees left in the performance of the test. From zero position on the center line, elevate the test fixture 500 mils. Revolve the mount by means of the elevation knob 500 mils in a counter direction and sight on the center vertical line. During this operation, the optical line of sight shall not deviate off the line by more than 0.5 mil. Observe that the horizontal reticle line of the telescope is in coincidence with the horizontal line of the target image. Repeat this procedure with the test fixture set at angles of 800 mils and 1100 mils. Starting from zero, cant the mount 7 degrees to the left of the vertical position utilizing the cross leveling handwheel of the fixture. An angle of 500 mils shall be set in the test fixture. Return the canted mount to a leveled position utilizing the cross level knob and center the cross level vial bubble, bringing the optical line of sight of the telescope in coincidence with the left vertical (plumb) line, and

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rotate the elevation knob to bring the line of sight in coincidence with the 500 mil elevation angle mark on the target. Note the amount of movement on the elevation micrometer knob scale. Repeat this procedure for elevation angles of 800 and 1100 mils, and note azimuth correction readings. Repeat the azimuth correction readings with the test fixture canted 7 degrees right of the vertical position, starting with elevation angles from 1100 to 500 mils. After testing, set the fixture at zero elevation and at zero cross leveling angle. Then level the mount in both directions. The established vertical (center) line shall be in coincidence with the optical line of sight of the telescope. All azimuth correction readings shall conform to Table II and 3.7. The plumb travel test of 4.5.10 may be performed concurrently with the test above. The following tests may be performed utilizing the set up and target as outlined above.

4.5.5 Elevation error.- The mount shall be positioned as specified in 4.5.3, with the auxiliary telescope sighted on the target in 4.5.4. The telescope reticle center shall be observed and noted with respect to the target's graduations. The elevation fixture shall then be depressed to indicate the angle of 1200 mils. The mount shall then be elevated by means of the elevation knob to the position indicating 1200 mils in elevation. This will position the telescope's reticle center coincident with the target's center within the limits specified in 3.5.1. The elevation fixture shall then be elevated at 100 mil increments and the mount shall be depressed at 100 mil increments. Any error in elevation through the full excursion of 1200 mils shall be noted and shall not exceed the specified limits.

4.5.6 Backlash-elevation mechanism.- Elevation backlash shall be checked at the 100 mil points, without overtravel, as outlined herein. The telescope's reticle center shall be observed at each point and noted with respect to target. The mount's elevation knob shall then be rotated in a forward direction an additional 100 mils, then returned to its former position without overtravel. The elevation backlash error shall be read directly on the elevation micrometer scale and shall not exceed 0.5 mil as specified in 3.5.3(a).

4.5.7 Angle of site error.- With the mount positioned as specified in 4.5.5, displace the angle of site level by setting the angle of site scale to the 100 mil graduation. By means of the elevation fixture, return the level bubble to its level position. Any difference in the angle set off by the angle of site mechanism and the angle set off by the elevation fixture in returning the level bubble to its level position shall not exceed the requirement specified in 3.5.2. Repeat this test in 100 mil increments through the excursion of 0 to 600 mils.

4.5.8 Backlash-angle of site mechanism.- To determine compliance with 3.5.3(b), the angle of site mechanism shall be checked at the 100 mil points, without overtravel, as specified herein. The line of sight shall be noted at each point with respect to the target. The mechanism shall then be advanced 100 mils and returned to its former position with-

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out overtravel. Backlash shall be read directly from the angle of site micrometer and shall be within the limit specified in 3.5.3(b).

4.5.9 Backlash-cross level mechanism.- The mount shall be positioned as specified in 4.5.5. Cross level the mount by means of the cross level worm knob until the cross level bubble is centralized. Rotate the mount through 800 mils of elevation and simultaneously rotate the elevation fixture an equal amount in the opposite direction. Observe that the position of the level bubble is centralized at each 100 mil forward rotation and return to original position. The backlash error of the cross level mechanism shall be read directly on the target. The error shall be within the limits specified in 3.5.3(c).

4.5.10 Plumb travel.- The mount shall be positioned as specified in 3.5.4 and 4.5.3 with the plumb line on the target and the auxiliary telescope's vertical reticle line in coincidence. The elevation fixture shall then be depressed at 100 mil increments while simultaneously operating the mount a like number of mils in the opposite direction. At each step, sight through the telescope at the vertical target line. Any deviation in plumb travel can be read directly on the target and shall be within the limits specified in 3.5.4.

4.5.11 Cant.- The mount shall be positioned as specified in 3.5.5 and 4.5.3 and the line of sight through the telescope reticle center shall be in coincidence with the target's center after canting the locating surface of the mount 7 degrees to the left by means of the elevation test fixture. The position of the reticle lines shall be observed and noted with respect to the target graduations. When this is established, perform the elevation travel test specified in 4.5.5. Any deviation of the position of the reticle cross lines with the target's graduations shall not exceed the requirements specified in 3.5.5. Repeat the test with the mount canted 7 degrees in the opposite direction. Any deviation shall not exceed the limits specified.

4.5.12 Operability (torque).- The running torque used to rotate the knobs listed in Table I and 3.6.1 shall be tested by means of standard torque measuring equipment and suitable adapters. The indicated readings shall be within the limits in Table I at the standard ambient temperature (60° to 90°F). This test shall also be performed when the mount is thermally stabilized at the extreme temperature conditions during the temperature test specified in 4.5.1. The measured values shall not exceed the limits specified in Table I.

4.5.12.1 Elevation lock.- With the elevation lock secured, apply a torque of 30 in. lb. to the elevation knob. The measured torque specified in 3.6.1 shall not loosen or permit rotation of the knob, to insure compliance with 3.6.1.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing, and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P7597736. The level of protection shall be as specified in the procurement document.

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6. NOTES

6.1 Intended use.- Telescope Mount: M3A1 is used as a component part of Sightunits: M23, M28A1, M29, and M29A1 and as a support for the Panoramic Telescope: M1.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation, packaging, and packing (5.1).
(c) Applicable packaging data sheet number (5.1).
(d) Applicable stock number.

Custodian:

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Preparing activity:

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